

Meeting 2 — Total Differentials, Exact Forms, and Integration Along Curves

Mathematics for Chemists | 2.5–3 hours

Guiding question for the whole meeting:

“When does a physical quantity depend only on where you are, and not on how you got there?”

This question runs through every topic today. The answer will connect partial derivatives, thermodynamic potentials, path integrals, and — in the final section — the topology of the plane itself.

Part 1 — The Total Differential (~30 min)

1.1 Linearising a Multivariable Function

In Meeting 1 we found critical points by setting $\nabla f = 0$. But the gradient contains more information: it tells us *how fast* f changes in every direction. Recall the quadratic approximation at a critical point:

$$f(x_0 + \mathbf{h}) \approx f(x_0) + \frac{1}{2} \mathbf{h}^T H \mathbf{h}.$$

Away from critical points, the dominant term is linear:

$$f(x_0 + \mathbf{h}) \approx f(x_0) + \nabla f(x_0) \cdot \mathbf{h}.$$

We now package this linear approximation into a concise notation.

Definition. For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ continuously differentiable, the **total differential** of f at a point x is the expression

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n.$$

Here $dx_1, \dots, dx_n$ are symbolic displacements (formal placeholders for small changes in each variable). The total differential encodes the best linear approximation to Δf : given a small displacement $(\Delta x_1, \dots, \Delta x_n)$,

$$\Delta f = f(x + \Delta x) - f(x) \approx \frac{\partial f}{\partial x_1} \Delta x_1 + \cdots + \frac{\partial f}{\partial x_n} \Delta x_n = df|_{(\Delta x_1, \dots, \Delta x_n)}.$$

The error is $o(\|\Delta x\|)$: smaller than any linear term as $\|\Delta x\| \rightarrow 0$.

What df is not. The symbols dx_i are *not* “infinitely small numbers”. They are the coordinates on the tangent space at x — formal directions. The expression df is a **linear map**: it takes a displacement vector and returns a number. We will not call it a “differential form” in this course, but this is exactly what it is.

Example 1. $f(x, y) = x^2 + y^2$:

$$df = 2x dx + 2y dy.$$

Numerical check at $(1, 1)$ with $\Delta x = 0.1$, $\Delta y = 0.2$: $\Delta f = (1.1)^2 + (1.2)^2 - 2 = 0.65$; $df = 2(1)(0.1) + 2(1)(0.2) = 0.6$. Error: 0.05, of order $\|\Delta x\|^2$.

Example 2. $f(x, y, z) = xyz$:

$$df = yz dx + xz dy + xy dz.$$

If we move only in the x -direction ($dy = dz = 0$): $df = yz dx$. Dividing by dx recovers $\partial f/\partial x = yz$. The total differential contains all partial derivatives as special cases.

1.2 The Chain Rule as Total Differential

Suppose $(x_1(t), \dots, x_n(t))$ is a smooth curve in $\mathbb{R}^n$ and f is evaluated along it. Dividing df by dt :

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \frac{dx_1}{dt} + \dots + \frac{\partial f}{\partial x_n} \frac{dx_n}{dt} = \nabla f \cdot \dot{\mathbf{x}}.$$

This is the multivariable chain rule, and df is the natural way to state it without choosing a particular parameter. The total differential df is, precisely, the thing that becomes df/dt when we restrict to a parametrised curve.

Example 3. The equilibrium constant satisfies $\ln K(T) = -\Delta G^\circ(T)/RT$ where $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ (approximately constant $\Delta H^\circ, \Delta S^\circ$). Computing $d(\ln K)/dT$:

$$\frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta G^\circ/T)}{dT} = -\frac{1}{R} \cdot \frac{-\Delta H^\circ}{T^2} = \frac{\Delta H^\circ}{RT^2}.$$

This is the **van't Hoff equation**: an exothermic reaction ($\Delta H^\circ < 0$) has K decreasing with T , and vice versa. It follows from nothing but the chain rule applied to a function of one variable.

Part 2 — Thermodynamic Potentials (~35 min)

2.1 The Fundamental Relation

Combining the first law ($dU = \delta Q - p dV$) and the second law ($\delta Q_{\text{rev}} = T dS$) for a reversible process in a closed system doing only pV -work:

$$dU = T dS - p dV.$$

This is the total differential of U viewed as a function $U(S, V)$. Reading off coefficients:

$$\left(\frac{\partial U}{\partial S}\right)_V = T, \quad \left(\frac{\partial U}{\partial V}\right)_S = -p.$$

Nothing is assumed beyond smoothness of U . The natural variables of U are (S, V) — the variables appearing in dU .

The practical problem. Experiments are rarely done at constant S and V . We need potentials whose natural variables are T , p , or combinations. The tool is the Legendre transform: adding a term $d(xy) = x dy + y dx$ to swap which variable is “free”.

2.2 The Four Potentials by Legendre Transform

Enthalpy $H = U + pV$ (replace $V \rightarrow p$):

$$dH = dU + p dV + V dp = T dS + V dp.$$

Helmholtz free energy $F = U - TS$ (replace $S \rightarrow T$):

$$dF = dU - T dS - S dT = -S dT - p dV.$$

Gibbs free energy $G = U - TS + pV$ (replace both):

$$dG = -S dT + V dp.$$

Potential	Definition	$d\phi$	Natural vars	Partial derivatives
U	—	$T dS - p dV$	S, V	$T = \left(\frac{\partial U}{\partial S}\right)_V, -p = \left(\frac{\partial U}{\partial V}\right)_S$
H	$U + pV$	$T dS + V dp$	S, p	$T = \left(\frac{\partial H}{\partial S}\right)_p, V = \left(\frac{\partial H}{\partial p}\right)_S$
F	$U - TS$	$-S dT - p dV$	T, V	$-S = \left(\frac{\partial F}{\partial T}\right)_V, -p = \left(\frac{\partial F}{\partial V}\right)_T$
G	$U - TS + pV$	$-S dT + V dp$	T, p	$-S = \left(\frac{\partial G}{\partial T}\right)_p, V = \left(\frac{\partial G}{\partial p}\right)_T$

2.3 Maxwell Relations from Schwarz's Theorem

Each potential is a smooth function, so its mixed second partial derivatives are equal (Schwarz's theorem). Applied to $G(T, p)$:

$$\frac{\partial^2 G}{\partial p \partial T} = \frac{\partial^2 G}{\partial T \partial p} \Rightarrow \frac{\partial}{\partial p}(-S) = \frac{\partial}{\partial T}(V) \Rightarrow -\left(\frac{\partial S}{\partial p}\right)_T = \left(\frac{\partial V}{\partial T}\right)_p.$$

Potential	$d\phi = M dx + N dy$	Maxwell relation: $\partial M / \partial y = \partial N / \partial x$
$U(S, V)$	$T dS - p dV$	$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$
$H(S, p)$	$T dS + V dp$	$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$
$F(T, V)$	$-S dT - p dV$	$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$
$G(T, p)$	$-S dT + V dp$	$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$

Every Maxwell relation is one application of Schwarz's theorem to one thermodynamic potential. They allow entropy derivatives (unmeasurable) to be replaced by p, V, T derivatives (measurable).

2.4 Derivation of $C_p - C_V$

Step 1. $C_V = T(\partial S / \partial T)_V$, $C_p = T(\partial S / \partial T)_p$, so $C_p - C_V = T[(\partial S / \partial T)_p - (\partial S / \partial T)_V]$.

Step 2. Treat $S = S(T, V)$. Divide $dS = (\partial S / \partial T)_V dT + (\partial S / \partial V)_T dV$ by dT at constant p :

$$\left(\frac{\partial S}{\partial T}\right)_p = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p.$$

Step 3. Apply the Maxwell relation from F : $(\partial S/\partial V)_T = (\partial p/\partial T)_V$. Then:

$$C_p - C_V = T \left(\frac{\partial p}{\partial T} \right)_V \left(\frac{\partial V}{\partial T} \right)_p.$$

Step 4. With $\alpha = \frac{1}{V}(\partial V/\partial T)_p$ and $\kappa_T = -\frac{1}{V}(\partial V/\partial p)_T$, the cyclic identity gives $(\partial p/\partial T)_V = \alpha V/\kappa_T$, and:

$$C_p - C_V = \frac{TV\alpha^2}{\kappa_T}.$$

Since $\kappa_T > 0$ always, $C_p \geq C_V$. For an ideal gas $\alpha = 1/T$, $\kappa_T = 1/p$, recovering $C_p - C_V = nR$.

Part 3 — Exact and Non-Exact Expressions (~35 min)

3.1 The Exactness Question

Consider an expression $P(x, y) dx + Q(x, y) dy$. We ask: is this the total differential df of some function f ? That is, does there exist f such that $\partial f/\partial x = P$ and $\partial f/\partial y = Q$?

If such f exists, the expression is called **exact**. If not, it is **non-exact**.

The necessary condition. If $P dx + Q dy = df$, then by Schwarz:

$$\frac{\partial P}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial Q}{\partial x}.$$

So $\partial P/\partial y = \partial Q/\partial x$ is **necessary** for exactness.

Theorem (exactness criterion)

On a *simply connected* domain $U \subseteq \mathbb{R}^2$, the expression $P dx + Q dy$ is exact if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \quad \text{everywhere on } U.$$

The condition “simply connected” means the domain has no holes. We will see in Part 5 that holes can destroy the sufficiency.

Examples.

- $y dx + x dy$: $\partial P/\partial y = 1 = \partial Q/\partial x$. Exact. Indeed $y dx + x dy = d(xy)$.
- $y dx - x dy$: $\partial P/\partial y = 1$, $\partial Q/\partial x = -1$. Not equal: **non-exact**. No function f has $\partial f/\partial x = y$ and $\partial f/\partial y = -x$ simultaneously.
- $2xy dx + (x^2 + e^y) dy$: $\partial P/\partial y = 2x = \partial Q/\partial x$. Exact.

3.2 Finding the Potential

Given an exact expression $P dx + Q dy$, find f .

Method.

1. Integrate P with respect to x : $f(x, y) = \int P(x, y) dx + g(y)$, where $g(y)$ is an unknown function of y alone.

2. Differentiate with respect to y and set equal to Q : $\partial f/\partial y = (\text{known}) + g'(y) = Q(x, y)$.
3. Solve for $g'(y)$ and integrate.

The answer is unique up to an additive constant.

Example. $\omega = 2xy dx + (x^2 + e^y) dy$. Step 1: $f = x^2y + g(y)$. Step 2: $\partial f/\partial y = x^2 + g'(y) = x^2 + e^y$, so $g'(y) = e^y$, $g(y) = e^y$. Result: $f(x, y) = x^2y + e^y$. Check: $\partial f/\partial x = 2xy$, $\partial f/\partial y = x^2 + e^y$. $\square$

3.3 Heat and Work: Non-Exact Expressions in Thermodynamics

In thermodynamics, the notation δ signals non-exactness:

$$\delta Q = T dS \quad (\text{heat absorbed}), \quad \delta W = p dV \quad (\text{work done by system}).$$

There is no “total heat function” $Q(S, V)$ whose differential is $T dS$. The heat absorbed in a process depends on the path from initial to final state, not just the endpoints.

Yet $dU = \delta Q - \delta W = T dS - p dV$ is exact: the path-dependences of Q and W cancel exactly, leaving a quantity that depends only on the thermodynamic state. The distinction between state functions (U , H , F , G , S) and path functions (Q , W) is precisely the distinction between exact and non-exact expressions.

Geometric picture. An exact expression df has level curves $f = c$ as its “equipotential surfaces”. The gradient ∇f is everywhere perpendicular to these surfaces. Non-exact expressions have no such structure: they do not point “uphill” on any single surface, because no surface exists.

Part 4 — Integration Along a Curve (~40 min)

4.1 Definition of the Line Integral

Let γ be a smooth curve in $\mathbb{R}^2$, parametrised by $(x(t), y(t))$ for $t \in [a, b]$. The **line integral** of $P dx + Q dy$ along γ is:

$$\int_{\gamma} P dx + Q dy = \int_a^b \left(P(x(t), y(t)) \frac{dx}{dt} + Q(x(t), y(t)) \frac{dy}{dt} \right) dt.$$

This reduces everything to an ordinary definite integral. The result may depend on the path γ , not just its endpoints.

Example 1. Compute $\int_{\gamma} y dx + x dy$ from $(0, 0)$ to $(1, 1)$.

Path (a): straight line $x = t$, $y = t$, $t \in [0, 1]$:

$$\int_0^1 (t \cdot 1 + t \cdot 1) dt = \int_0^1 2t dt = 1.$$

Path (b): parabola $x = t$, $y = t^2$, $t \in [0, 1]$:

$$\int_0^1 (t^2 \cdot 1 + t \cdot 2t) dt = \int_0^1 3t^2 dt = 1.$$

Both give 1. Why? Because $y dx + x dy = d(xy)$ is exact.

4.2 The Fundamental Theorem for Line Integrals

Theorem (Fundamental theorem for line integrals)

If $P dx + Q dy = df$ on a domain U , then for any smooth curve $\gamma \subset U$ from point A to point B :

$$\int_{\gamma} P dx + Q dy = f(B) - f(A).$$

Proof. Parametrise γ by $(x(t), y(t))$, $t \in [a, b]$. Then:

$$\int_{\gamma} df = \int_a^b \frac{df}{dt} dt = \int_a^b \frac{d}{dt} f(x(t), y(t)) dt = f(x(b), y(b)) - f(x(a), y(a)) = f(B) - f(A). \quad \square$$

Consequences.

- The integral depends only on the endpoints, not the path: **path independence**.
- For any *closed* loop ($A = B$): $\oint_{\gamma} df = 0$.
- Conversely, if $\oint_{\gamma} P dx + Q dy = 0$ for every closed loop in U , then the expression is exact on U .

These three properties (exactness, path independence, vanishing on closed loops) are equivalent on simply connected domains.

This is the exact multivariable analogue of the fundamental theorem of calculus: $\int_a^b f'(x) dx = f(b) - f(a)$.

4.3 Chemistry: Thermodynamic Work Along a Process

The work done by a gas along a path γ in the (V, p) -plane is $W = \int_{\gamma} p dV$. Since $p dV$ is non-exact, W depends on the path.

Isothermal expansion of an ideal gas. Along the isotherm $p = nRT/V$ from V_1 to V_2 :

$$W = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln \frac{V_2}{V_1}.$$

Two-step process. First expand at constant pressure p_1 from V_1 to V_2 ; then cool at constant volume V_2 back to the original T :

$$W = p_1(V_2 - V_1) + 0 = p_1(V_2 - V_1) \neq nRT \ln \frac{V_2}{V_1}.$$

The same initial and final states, different work.

By contrast, $\Delta U = \int_{\gamma} dU$ depends only on the endpoints: this is the first law of thermodynamics, restated as the exactness of dU .

4.4 A Non-Exact Example

Compute $\int_{\gamma} \frac{-y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy$ around the unit circle $x = \cos t$, $y = \sin t$, $t \in [0, 2\pi]$.

Substituting: $dx = -\sin t dt$, $dy = \cos t dt$, $x^2 + y^2 = 1$:

$$\int_0^{2\pi} [(-\sin t)(-\sin t) + (\cos t)(\cos t)] dt = \int_0^{2\pi} 1 dt = 2\pi \neq 0.$$

The integral around a closed loop is nonzero. Call this expression ω . Despite the exactness test $\partial P/\partial y = \partial Q/\partial x$ holding wherever it is defined, ω is not exact. The resolution requires understanding the *topology* of the domain.

Exercises (~25 min)

Exercise 1 (chain rule). Compute df for $f(x, y) = e^{xy} \sin y$. Then, for the path $(x(t), y(t)) = (t, \pi t)$, compute df/dt using the chain rule formula.

Exercise 2 (van't Hoff). Starting from $\ln K = -\Delta G^\circ/RT$ and $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ with $\Delta H^\circ, \Delta S^\circ$ constant, derive $d(\ln K)/dT = \Delta H^\circ/RT^2$.

Exercise 3 (Maxwell relation). Starting from $dF = -S dT - p dV$, read off the partial derivatives of F and derive the Maxwell relation $(\partial S/\partial V)_T = (\partial p/\partial T)_V$.

Exercise 4 (exactness and potential). Test whether each of the following is exact. For those that are, find the potential f .

- (a) $3x^2y dx + x^3 dy$
- (b) $(2xy - \sin x) dx + (x^2 + e^y) dy$
- (c) $y dx - x dy$
- (d) $(yz + z) dx + xz dy + (xy + x) dz$ (check all three mixed-partial conditions)

Exercise 5 (line integrals). Compute $\int_\gamma x dy - y dx$ along: (a) the upper semicircle from $(-1, 0)$ to $(1, 0)$; (b) the lower semicircle from $(-1, 0)$ to $(1, 0)$. Do the results agree? Why or why not? (*Hint: compute the exactness condition first.*)

Exercise 6 (thermodynamic cycle). An ideal gas undergoes a cycle: (i) isothermal expansion at temperature T_1 from V_1 to $2V_1$; (ii) isochoric cooling from T_1 to T_2 ; (iii) isothermal compression at T_2 back to V_1 ; (iv) isochoric heating back to T_1 . Compute the total work $\oint p dV$ done in the cycle. Why is $\oint dU = 0$ regardless?

Exercise 7 (finding the constant a). The expression $(axy^2 + by) dx + (x^2y + bx + cy^2) dy$ is exact for certain values of a, b, c . Find all such values and, for those values, find the potential.

Exercise 8 (conservative fields). The force field $\mathbf{F} = (xy, x^2)$ acts on a particle. Show that $\mathbf{F}$ is not conservative by testing the exactness condition. Verify by computing the work along two different paths from $(0, 0)$ to $(2, 1)$: the straight line, and the path along the x -axis followed by a vertical segment.

Part 5 — Advanced: Topology of the Punctured Plane (demonstration, not examinable)

Everything from here onward is a demonstration. The goal is to show where the mathematics of this meeting leads — to some of the deepest ideas in 20th-century mathematics and to measurable physical effects. No proofs are required from students.

6.1 Act 1 — A Paradox

We proved: on a simply connected domain, if $\partial P/\partial y = \partial Q/\partial x$, then $\oint_\gamma P dx + Q dy = 0$ for every closed loop. Now consider:

$$\omega = \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

Compute:

$$\frac{\partial P}{\partial y} = \frac{-(x^2 + y^2) + 2y^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad \frac{\partial Q}{\partial x} = \frac{(x^2 + y^2) - 2x^2}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2}.$$

The condition $\partial P/\partial y = \partial Q/\partial x$ is satisfied wherever ω is defined.

Yet we computed $\oint_{\text{unit circle}} \omega = 2\pi \neq 0$.

Write $2\pi \neq 0$ on the board. Wait.

6.2 Act 2 — The Diagnosis

The proof of the exactness theorem constructs a potential $f(x, y) = \int_{(x_0, y_0)}^{(x, y)} P dx + Q dy$ by integrating from a fixed base point along a path. If the domain has a hole, two paths with different winding around the hole can give different values of f : the “potential” is not single-valued.

Where is ω undefined? At $(0, 0)$: the denominators vanish. The domain is $\mathbb{R}^2 \setminus \{(0, 0)\}$ — the **punctured plane**. This domain has a hole, and the unit circle winds once around it.

In polar coordinates: a short computation shows $\omega = d\theta$, where $\theta = \arctan(y/x)$ is the polar angle. The angle θ is a perfectly good local function near any point away from the origin — but it is *not* a global function on the punctured plane. Going once around the origin, θ increases by 2π and does not return to its starting value. This is why no global potential exists: the potential *would be* θ , but θ is multi-valued.

6.3 Act 3 — The Winding Number

Definition. The **winding number** $n(\gamma, 0)$ of a closed curve γ around the origin is the number of times γ goes around the origin counterclockwise (negative if clockwise).

Draw on board: a circle ($n = 1$), a circle traversed twice ($n = 2$), a figure-eight not enclosing the origin ($n = 0$), a clockwise circle ($n = -1$).

For any closed curve γ in $\mathbb{R}^2 \setminus \{0\}$:

$$\oint_{\gamma} \omega = 2\pi n(\gamma, 0).$$

The winding number is a **topological invariant**: it does not change under continuous deformation of γ , as long as γ never crosses the origin. This is why it measures something intrinsic to the topology of the domain, not to the particular path chosen.

6.4 Act 4 — Connection to Complex Analysis

Identify (x, y) with $z = x + iy \in \mathbb{C}$. Consider the complex expression:

$$\frac{dz}{z} = \frac{dx + i dy}{x + iy}.$$

Multiplying numerator and denominator by $\bar{z} = x - iy$:

$$\frac{dz}{z} = \frac{x dx + y dy}{x^2 + y^2} + i \frac{x dy - y dx}{x^2 + y^2} = d(\ln r) + i \omega,$$

since $d(\ln r) = (x dx + y dy)/r^2$ is exact. Therefore:

$$\oint_{\gamma} \frac{dz}{z} = 0 + i \cdot 2\pi n(\gamma, 0) = 2\pi i n(\gamma, 0).$$

This is the simplest case of the **residue theorem**. The “potential” of dz/z would be $\ln z = \ln r + i\theta$, the complex logarithm — which is multi-valued on $\mathbb{C} \setminus \{0\}$ precisely because θ increases by 2π around the origin.

The residue theorem in general. For a meromorphic function $f(z)$ with poles at $z_1, \dots, z_n$, each pole at z_k contributes a “non-exact part” proportional to $\omega_{z_k} = \text{Im}(dz/(z - z_k))$. The residue theorem states:

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_k n(\gamma, z_k) \cdot \text{Res}(f, z_k).$$

This is the statement that every non-exact closed form on the n -times punctured plane is a linear combination of the n generators ω_{z_k} , weighted by residues. The residue at a pole is the coefficient of the non-exact part of $f dz$ near that pole.

The power of complex analysis arises from the topology of punctured subsets of the plane. Integration along curves detects the holes.

6.5 Act 5 — Other Forms That Detect Topology

The form ω is not special: it is the first member of an infinite family. Each dimension has its own generator, detecting a different type of hole.

Dimension 2: the punctured plane. $\omega = d\theta$ on $\mathbb{R}^2 \setminus \{0\}$, detecting winding around a point.

Dimension 3: the punctured space. On $\mathbb{R}^3 \setminus \{0\}$, the 2-form:

$$\Omega = \frac{x dy \wedge dz - y dx \wedge dz + z dx \wedge dy}{r^3} = \frac{\mathbf{r}}{r^3} \cdot d\mathbf{S}$$

is closed but not exact: integrating Ω over any closed surface enclosing the origin gives 4π (the solid angle of the full sphere), regardless of the shape of the surface. This is the solid angle element.

Physically, Ω is — up to constants — the electric field of a point charge. The statement $\iint_S \Omega = 4\pi$ is **Gauss’s law**: the flux of $\mathbf{E}$ through any closed surface depends only on whether it encloses the charge, not on its shape. Gauss’s law is the statement that Ω is non-exact on the space with the charge removed.

Space	Hole	Form	Degree	Physics
$\mathbb{R}^2 \setminus \{0\}$	point	$d\theta$	1	Solenoid vector potential
$\mathbb{R}^3 \setminus \{0\}$	point	$\mathbf{r}/r^3 \cdot d\mathbf{S}$	2	Coulomb field; Dirac monopole
$\mathbb{R}^3 \setminus \{\text{line}\}$	line	$d\theta$ (cylindrical)	1	Magnetic field of wire; Aharonov–Bohm
Torus T^2	two loops	$d\theta_1, d\theta_2$	1, 1	Crystal momentum (Brillouin zone)
Parameter space	degeneracy	Berry connection $\mathbf{A}_n$	1	Berry phase; conical intersections

6.6 Act 6 — Physical Consequences of Topology

The Aharonov–Bohm effect (1959). An infinitely long solenoid carries magnetic flux Φ but has zero magnetic field $\mathbf{B} = 0$ outside it. The vector potential outside is:

$$\mathbf{A} = \frac{\Phi}{2\pi} \left(\frac{-y}{x^2 + y^2}, \frac{x}{x^2 + y^2} \right) = \frac{\Phi}{2\pi} \omega.$$

An electron travelling around the solenoid along a path γ acquires a quantum phase:

$$\phi = \frac{e}{\hbar} \oint_{\gamma} \mathbf{A} \cdot d\mathbf{r} = \frac{e\Phi}{2\pi\hbar} \cdot 2\pi n(\gamma, 0) = \frac{e\Phi n}{\hbar}.$$

This phase is measurable in interference experiments, even though the electron never enters a region where $\mathbf{B} \neq 0$. The physics is carried entirely by the topology of the accessible region (the punctured plane, with the solenoid excluded) and by the winding number of the electron's path. The experiment was confirmed in 1960.

The Dirac quantisation condition. A magnetic monopole at the origin produces $\mathbf{B} = g\mathbf{r}/r^3$. We need a vector potential $\mathbf{A}$ with $\nabla \times \mathbf{A} = \mathbf{B}$, but $\Omega = \mathbf{r}/r^3 \cdot d\mathbf{S}$ is non-exact: no globally defined $\mathbf{A}$ exists. Dirac's resolution: define $\mathbf{A}$ on two overlapping patches (northern and southern hemispheres) and require the gauge transformation relating them on the equator to be single-valued. This forces:

$$\frac{eg}{\hbar c} \in \mathbb{Z}.$$

A topological constraint forces a physical quantity to be quantised. This is the deepest kind of argument in theoretical physics: no dynamics, no approximation — pure topology.

The Berry phase and conical intersections. When a quantum Hamiltonian $H(\mathbf{R})$ depends on slowly varying parameters $\mathbf{R}$ that are cycled around a closed loop $\mathcal{C}$ in parameter space, an eigenstate $|n(\mathbf{R})\rangle$ acquires a geometric phase:

$$\gamma_n = \oint_{\mathcal{C}} \mathbf{A}_n \cdot d\mathbf{R}, \quad \mathbf{A}_n = i\langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle.$$

The 1-form $\mathbf{A}_n \cdot d\mathbf{R}$ (the **Berry connection**) is non-exact when parameter space has a hole — typically a point where two energy levels become degenerate (a **conical intersection**).

In molecular physics: when nuclear coordinates cycle around a conical intersection of two Born–Oppenheimer surfaces, the electronic wavefunction acquires a Berry phase of π . This modifies the vibronic spectrum and reaction dynamics and has been observed in molecules such as Na_3 and in the photochemistry of cyclopentadienyl radical. The conical intersection is the puncture; the Berry phase is the winding integral.

6.7 Act 7 — The Unifying Framework

All of the above is unified by a single construction. For any smooth domain U , define:

- $Z^k(U)$: the space of closed k -forms ($d\omega = 0$)
- $B^k(U)$: the space of exact k -forms ($\omega = d\eta$)

Since $d(d\eta) = 0$ always ($d^2 = 0$), we have $B^k \subseteq Z^k$. The **k -th de Rham cohomology group** is:

$$H_{\text{dR}}^k(U) = Z^k(U)/B^k(U).$$

Its dimension b_k (the **k -th Betti number**) counts the independent “holes” of dimension k in U :

Domain U	b_1	Interpretation
$\mathbb{R}^2$	0	No holes; every closed form is exact
$\mathbb{R}^2 \setminus \{0\}$	1	One hole; ω generates H^1
$\mathbb{R}^2 \setminus \{z_1, \dots, z_n\}$	n	n holes; $\omega_{z_1}, \dots, \omega_{z_n}$ generate H^1
Torus T^2	2	Two independent loops

The **de Rham theorem** states that $H_{\text{dR}}^k(U) \cong H^k(U; \mathbb{R})$, the k -th singular cohomology of U — a purely topological object defined without any calculus. Differential forms, which are analytic objects, compute topological invariants.

The message of this entire section:

*Integration of differential forms is the correct way to probe the topology of a space.
Closed-but-not-exact forms are topological detectors — one for each type of hole and each dimension.*

The form ω we computed today is the simplest case. The theory continues into every dimension, and its echoes appear in Gauss's law, the Aharonov–Bohm effect, the Berry phase, topological insulators, and the classification of gauge theories. Students wishing to pursue this should look at Bott and Tu, *Differential Forms in Algebraic Topology* (Springer, 1982) — which begins with exactly the example computed today.

Summary and Bridge to Meeting 3

What we did today.

- The total differential $df = \sum_i (\partial f / \partial x_i) dx_i$ linearises a multivariable function and encodes the chain rule.
- The four thermodynamic potentials arise by Legendre transforms of $U(S, V)$; their fundamental relations are total differentials; Maxwell relations follow from Schwarz's theorem.
- An expression $P dx + Q dy$ is exact ($= df$) iff $\partial P / \partial y = \partial Q / \partial x$ on a simply connected domain. Non-exact expressions appear as heat and work.
- The line integral $\int_\gamma P dx + Q dy$ is path-independent iff the expression is exact; for exact forms, $\int_\gamma = f(B) - f(A)$.
- On the punctured plane, the form $\omega = d\theta$ is closed but not exact; $\oint_\gamma \omega = 2\pi n(\gamma, 0)$. This detects topology.

Looking ahead — Meeting 3: Coordinate Systems and the Implicit Function Theorem. Today we integrated along curves in the plane using Cartesian coordinates. In Meeting 3 we will ask: what if the natural coordinates are polar, cylindrical, or spherical? The change of variables formula requires the Jacobian determinant — the same determinant that appears in the implicit function theorem and that vanishes at coordinate singularities. The unifying theme will again be: *when is a smooth map locally invertible?*